

VideoSQL: Ultra Long Video Understanding by Agentic SQL Queries with Trainable Skills

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Abstract

1. Introduction

Ultra long video understanding (ULVU)

2. Related Works

2.1. Ultra Long Video Understanding

2.2. SQL-based RAG

Relational databases have long been the default for storing structured data. Recent work starts applying them to agentic RAG and agent memory. SAG [10] argues that naive RAG yields shallow retrieval, while graph-based RAG is slow, costly, and hard to update. They therefore use SQL as the backbone of RAG. Documents are stored as an event-entity incidence table. At query time, an LLM extracts entities from the query, and a JOIN retrieves events sharing these entities as seeds. A reverse JOIN then collects the entities attached to these seed events, and a further JOIN brings in previously unseen events incident to them. The expanded event set is passed back to the LLM, which selects the evidence and maps it to the original chunks. There are also other works that frame agent memories as databases like [7].

2.3. Skill Training

Training model weights, especially with reinforcement learning (RL), is expensive, slow, and not portable (you would need to host and run this specific model, which might perform poorly on general tasks). Many recent works began to train the agent's Skill document instead of model weights, which can be seen as a form of prompt optimization.

Training-free GRPO [1] uses an approach similar to the original gradient-based GRPO [8]. An initial experience and a policy model are used to roll out a batch of traces, and another LLM computes the semantic advantage of each

rollout, analogous to the numeric advantage in the original GRPO. These semantic advantages are used to update the experience. **Trace2Skill** [6] uses parallel rollouts in batches to collect many traces and generate patches for successful and failed traces. These patches are then merged into the final Skill in sequence. Compared to training-free GRPO, this can be seen as a form of offline reinforcement learning. **SkillOpt** [12] treats Skill optimization similarly to gradient descent, where the editing budget is analogous to the learning rate, the gradient direction is analogous to the editing direction, etc. It also learns a meta-skill to track what kind of edit is helpful using a held-out validation set. **SkillCAT** [3] uses DPO-style training. It samples two paired rollouts: a successful one (r^+) and a failed one (r^-). It then analyzes the two traces, finds the point where they diverge, and generates a patch contrastively. The patch is then verified to see whether it helps the agent, similar to the gate in SkillOpt, and is merged if it turns a failure into a success or maintains a success. However, in many cases, the two traces are completely independent (with no shared prefix beyond the prompt), or the divergence point does not affect the result.

3. VideoSQL

3.1. Indexing

For fair comparison with our baseline MAGIC-Video [5], we re-used their index. We convert their index from a graph data structure into a relational database by constructing a table for each node type, and use utter key as edge. We also formulated the topic chain and event chain into tables. This yield major tables of episodes captions, semantic triplets, entities, and topic and event chains. The detailed schema of our database is in appendix and readers can also refer to the MAGIC-Video paper for details of these indices.

3.2. SQL-based queries

MAGIC-Video uses a Personalized Page Rank (PPR) to retrieval nodes in the graph by a query generated by the LLM, which the agent has no control over the relations between nodes. We still keep a simple semantic search tool for

model to search all text and images using embeddings. For example, if the agents want to find who ate the apple yesterday noon, it likely issues a query like "person ate apple yesterday noon" and wish the PPR returns relevant nodes. Instead, we formulate this using a relational database. Given the same indices structure as MAGIC-Video, the agent can write a SQL query like in Listing 1. Even though searches like this could be replaced with dedicated tools like `search_semantic(object="apple", time_s=time_s, time_end=time_end)`, using SQL as an interface provides a unified interface with more freedom and does not require specific design for each case (for example, if we need to search "the person who ate the apple that is not luke", we would need to re-design the tool signature). It can also be seen as a form of Programmatic Tool Calling. Such interface can also handle complex queries using advanced SQL features like JOIN, GROUP BY, sub-queries when needed.

Our work differs from SAG [10] that the LLM controls the SQL generation and it can use all SQL features, unlike SAG which only used JOIN as a way to expand the seed nodes.

Listing 1. An example SQL query for finding who ate the apple within a given time window.

```
SELECT *
FROM semantic_triples
WHERE predicate IN ('eat', 'eats', 'ate',
, 'eating')
AND object IN ('apple', 'the_apple', '
an_apple')
AND snapshot_s >= $start_s
AND snapshot_s < $end_s;
```

3.3. Trainable Skills

4. Evaluations

4.1. Datasets

We used two most popular ULVU datasets in the paper: EgoLifeQA [11] and MM-Lifelong [2]. EgoLifeQA uses multiple-choice questions (MCQ) while MM-Lifelong uses open QA. EgoLifeQA contains one week of recording in the same setting while MM-Lifelong has 3 different settings (day, week, and month) each with its own video. The MM-Lifelong week subset has the same video as EgoLifeQA. We argue that MCQ does not reflect realistic usage since most questions asked by users in real world are open. Additionally, agents can brute-force verify each option to artificially boost accuracy. We therefore reformulated EgoLifeQA into open QA questions (Open-EgoLifeQA) similar to MM-Lifelong and evaluated both using LLM-as-judge for semantic comparison between the ground-truth and predicted answers. To compare with prior work, we used the

Jack view from EgoLifeQA. We evaluated our un-trained VideoSQL on both Open-EgoLifeQA and MM-Lifelong validation set.

To train the VideoSQL Skill, we need to select a independent training set. Ego-R1 Data [9] is a training set build on top of EgoLifeQA and authors of Ego-R1 trained their method on this dataset and then tested on EgoLifeQA. MM-Lifelong also has a training split as well. However, these training sets overlap with their corresponding validation/testing sets at the video level. While no questions are leaked, we argue that sharing videos could still inflate performance, or at least fail to test the model's ability for cross-domain generalization. To address these, we trained out Skill on EgoLifeQA and tested it on more challenge MM-Lifelong (For context, MAGIC-Video reached 67.6 on EgoLifeQA but only 24.5 on MM-Lifelong).

4.2. Baselines

We selected two baselines for the training free harness, MAGIC-Video [5] and GROVE [4], all using GPT-5.6-luna as policy model. This will be used to compare with our harness with untrained skills. We also selected Trace2Skill [6] and SkillOpt [12] for our skill training methods, both build on top of the same VideoSQL harness.

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